# A Theory of Slack

## How Economic Slack Shapes Markets, Business Cycles, and Policies

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# CHAPTER 1.

# Introducing slack into business cycle research

Economic research into business cycles began in earnest after the Great Depression, which started in 1929 in the United States. Early economists—beginning with Keynes—naturally concentrated on its most devastating aspect: mass unemployment. A quarter of the labor force was thrown into unemployment at the worst of the Great Depression. This had terrible consequences—vividly described by John Steinbeck (1939, pp. 317–321) in *The Grapes of Wrath*:

> And then the dispossessed were drawn west—from Kansas, Oklahoma, Texas, New Mexico; from Nevada and Arkansas families, tribes, dusted out, tractored out. Carloads, caravans, homeless and hungry; twenty thousand and fifty thousand and a hundred thousand and two hundred thousand. They streamed over the mountains, hungry and restless—restless as ants, scurrying to find work to do—to lift, to push, to pull, to pick, to cut—anything, any burden to bear, for food. The kids are hungry. We got no place to live. Like ants scurrying for work, for food, and most of all for land. . . . In the camps the word would come whispering, There's work at Shafter. And the cars would be loaded in the night, the highways crowded—a gold rush for work. At Shafter the people would pile up, five times too many to do the work. A gold rush for work. They stole away in the night, frantic for work.

Fifty years later, when the devastation of the depression had faded into memory, modern macroeconomics moved in a different direction: away from unemployment to focus on fluctuations in prices and quantities. But of course, unemployment surges remain

a central feature of recessions. They return every once in a while, even if we have forgotten about them.

So this book returns the focus to economic slack—a concept that encompasses unemployment as well as other unsold and idle resources. As we go along we will see how different types of slack matter in modern economies.

The book starts by reviewing essential facts about slack and the business cycle. It then presents a theory to explain why slack emerges and why it fluctuates. Finally, it uses the theory to derive principles for the stabilization of slack fluctuations using monetary and fiscal policy.

## 1.1. Slack in Keynes's General Theory

Perhaps even more than other disciplines, macroeconomics is the product of its time. Sitting in the early 1930s and trying to make sense of the Great Depression and offer advice to avoid such future economic calamities, Keynes (1936) wrote the *General Theory of Employment, Interest and Money*. This remains the most influential text in macroeconomics.

The *General Theory* offers a fairly balanced treatment of unemployment and inflation. It discusses inflation and unemployment to a significant extent, with slightly more coverage afforded to unemployment. In the book, inflation and deflation appear on 9 pages; unemployment is mentioned on 32 pages (table 1.1).

Although the *General Theory* does not cover forms of slack beyond unemployment, we know that slack was present in other forms during the Great Depression too: not only was it almost impossible for jobseekers to find jobs, but it was also almost impossible for shops to find customers and for factories to find buyers. As a result, many shops were empty, a lot of factories were idle, and these businesses eventually had to lay their workers off and shut down. As Temin (2000, p. 301) wrote about the Great Depression: "unemployment rose to a peak of 25 percent and stayed above 15 percent for the rest of the 1930s;" at the same time, "there were many idle economic resources in America for a full decade."

This book presents a theoretical framework which shows us that these two phenomena are intrinsically linked. Slack on the product market—the fact that firms cannot find customers because aggregate demand is low—generates slack on the labor market—the fact that workers cannot find jobs because labor demand is low. A key insight of the book is that slack on the product market generates slack on the labor market. In plain English, if firms cannot sell their production, they will not hire workers.

In fact, it is not just that the phenomena of product market and labor market slack are linked. It is that they can be modeled entirely symmetrically. So you do not need two different frameworks to think about them: the same theory can be applied to the labor market and product market, and both markets can be stacked to form a business cycle model with slack.

TABLE 1.1. Pages mentioning unemployment and inflation in the *General Theory* and New Keynesian texts, according to their indexes

| | Keynes (1936) | Gali (2008) | Woodford (2003) |
|---|---|---|---|
| Unemployment | 5–9, 13, 15, 16, 22, 109, 121, 128, 190, 235, 251, 274, 275, 278, 289, 334, 335, 346–349, 358, 359, 362–364, 381, 382 | 188 | |
| Inflation and related concepts | 119, 202, 207, 281, 301, 303, 291, 331, 332 | 1, 4, 11, 21–22, 31–32, 34, 37–38, 47, 49–52, 54–56, 60, 77, 79, 95–99, 103, 105, 107–108, 116, 120, 125–127, 130, 133–135, 137–139, 148, 155–156, 161–164, 168–176, 179, 189 | 3–5, 13–14, 17, 19–21, 39, 43–44, 46, 49, 90–101, 116–122, 126–138, 159–160, 176–177, 188, 203–205, 212–213, 215, 236, 248, 251, 262, 267, 272, 276–286, 289–295, 301–305, 317, 341, 347, 360–362, 381–382, 396–405, 408–409, 413–416, 418, 437, 440–441, 445–446, 460, 462–463, 467–485, 493, 499–501, 522–527, 544, 559–565, 576–582, 590–604, 615, 619–623, 639–641, 694–695, 714–715, 724–727 |
| Unemployment share | 78% | 2% | 0% |

Besides the term "inflation," Keynes (1936) also features "deflation." In Gali (2008), the terms related to inflation include "inflation dynamics" and "inflation targeting." In Woodford (2003), the terms related to inflation include "inflation inertia," "inflation rates," "inflation-targeting central banks," "inflation-targeting rules," and "deflation." The unemployment share is the share of pages mentioning unemployment in the total number of pages mentioning either unemployment or inflation and related terms.

## 1.2. Absence of slack in New Keynesian texts

While Keynes was writing during a mass-slack event, by the 1970s, the economic environment had changed, leading to drastic changes in macroeconomic theory. Following the fast economic growth of the US economy in the 1950s and 1960s, and the spike in inflation in the 1970s, the Old Keynesian model—which had formalized Keynes's ideas—was rapidly replaced by the Real Business Cycle model.

The Real Business Cycle model was built around Walrasian markets so it had no slack. Because it emphasized efficiency and monetary neutrality, it was not very useful for designing macroeconomic policies, and it was itself replaced by the New Keynesian model in the 1990s.

The New Keynesian model adopted the architecture of the Real Business Cycle model but swapped perfectly competitive markets for monopolistically competitive markets, which allowed researchers to introduce price rigidity and thus break efficiency and monetary neutrality. While the monopolistically competitive model produced rich price dynamics, it did not have room for unemployment or slack either, so the research squarely focused on inflation.

All in all, during the transition from Keynesian macroeconomics to New Keynesian macroeconomics, inflation was retained as a topic of interest but unemployment was lost. This omission of unemployment and slack and concentration on inflation appears clearly in New Keynesian textbooks. The leading New Keynesian texts are *Interest and Prices: Foundations of a Theory of Monetary Policy* by Woodford (2003) and *Monetary Policy, Inflation, and the Business Cycle* by Gali (2008). *Interest and Prices* discusses neither slack nor unemployment. *Monetary Policy, Inflation, and the Business Cycle* does not discuss slack, but unemployment is mentioned on one page at the end of the book, when possible extensions to the New Keynesian model are discussed.

Inflation, by contrast, is ubiquitous in both books (table 1.1). The term "inflation" appears on 97 pages in *Interest and Prices*. Additional related terms, such as "inflation inertia," "inflation-targeting central banks," and "inflation-targeting rules," appear on a further 104 pages. In total, there are 195 pages that cover inflation. In *Monetary Policy, Inflation, and the Business Cycle*, the term "inflation" appears on 42 pages. Related terms such as "inflation dynamics" and "inflation targeting" appear on 8 additional pages. In total there are 50 pages discussing inflation in that book.

## 1.3. Rebalancing business cycle theory

Policymakers in many countries have a dual legal mandate: keeping prices stable and maintaining the economy at full employment. Yet the New Keynesian model only speaks about price stability; it says nothing about full employment. The balanced treatment of

unemployment and inflation that was a hallmark of Keynesian thought disappeared in the New Keynesian literature, as the focus turned solely to inflation.

The book offers a new business cycle model that provides a more balanced treatment of inflation and unemployment—one that can help us think both about price stability and full employment. The model accounts for fluctuations not only in quantities and prices, but also in slack. A unilateral focus on inflation is problematic in the context of business cycles because slack varies far more than prices over the business cycle, and it carries substantial welfare costs. So slack must be taken into account to provide an accurate depiction of business cycles and relevant policy advice.

So what, exactly, is economic slack? Economic slack describes productive resources in the economy that are unsold and therefore idle. There are many different forms of slack: people who cannot find a job and remain unemployed; machines left idle in a factory; employed workers left idle on the job; hotel rooms, restaurant tables, airplane seats that remain vacant; durable goods that cannot be sold and depreciate; nondurable goods that cannot be sold and perish.

And why is it crucial to consider slack when designing business cycle stabilization policies? The main reason is that fluctuations in slack have large consequences for welfare. Economic slack represents a waste of productive resources, and it is therefore something that should be limited. In addition to its wastefulness, unemployment generates other, large costs to society. People who are unemployed suffer from lower mental and physical health than employed workers. Even employed people in areas with high unemployment report lower well-being. Accordingly, good economic policy should avoid periods of elevated slack.

The next question is: How does the book build a business cycle model with slack? The central assumption in the model is that both labor and product markets are organized around a matching function. The matching function gives the number of trades achieved when a certain number of buyers and sellers look for each other (Petrongolo and Pissarides 2001). Because matching takes time and effort, not all sellers are able to trade, which naturally generates slack in both labor and product markets. This differs from canonical business cycle models: the Old Keynesian model features nonclearing Walrasian markets; the Real Business Cycle model features Walrasian markets; and the New Keynesian model features monopolistic markets.

The book's central argument is that understanding markets and business cycles requires embracing a framework built around a matching function. By using a matching function, the framework is realistic: it features slack on all markets. The framework is also tractable: markets can be studied using a supply-demand apparatus that incorporates buyers' and sellers' trading probabilities. Such a slackish framework not only explains the sources and costly nature of unemployment, but it also provides a more accurate lens

through which to view markets, business cycles, and monetary and fiscal policy.

## 1.4. Overview of the book

Overall, the book is devoted to developing a new model of business cycles, and to exploring its implications for macroeconomic policy in the most transparent and practical way possible. The book is organized into five parts, each building on the last to develop the theory of slack and explore its implications.

Part I finishes setting the stage. Chapter 2 establishes the empirical foundation for the book, presenting essential facts about the existence, cyclicality, and social costs of slack that any business cycle model must address. For readers interested in the sequence of research that led to the framework developed here, chapter 3 recounts that journey. It details the real-world puzzles and policy debates that motivated each step. Readers eager to dive into the model might prefer to skip ahead.

Part II develops the theoretical heart of the book: a model of a generic slackish market. We begin in chapter 4 with the key building block—the matching function—before constructing a basic slackish market model in chapter 5. In the slackish market model, prices are set through norms. The price norm has critical implications for the behavior of the model; therefore, in chapter 6, we examine how prices are set in the real world to isolate the main empirical properties of the price norm. Then, in chapter 7, we insert a realistic, somewhat rigid price norm in the slackish market model and explore its properties.

After that, we explore several extensions and applications of the basic model and its efficiency properties. We explore how market capacity is determined (chapter 8), and how the model behaves in a dynamic setting, leading to the emergence of a Beveridge curve (chapter 9). We then define market efficiency, which provides a welfare benchmark against which to measure market outcomes (chapter 10). The part concludes by applying the slackish market model to the labor market (chapter 11). This is an important application because from a social perspective, unemployment is the most costly form of slack. There are also good data on labor market slack, which allows us to calibrate the model and compare its quantitative behavior with real-world patterns.

Part III scales up from a slackish market to a slackish economy. This part embeds the market model from part II into a macroeconomic framework to study business cycles. We start with a basic one-market economy in chapter 12 to understand the core properties of slackish business cycles. We then extend the framework to a more realistic economy with both labor and product markets (chapter 14). The two-market economy allows us to understand how slack percolates across markets. Finally, we incorporate a Phillips curve to link slack and inflation (chapter 13).

With a complete business cycle model in hand, part IV turns to the central policy questions. This part uses the slackish model to derive principles for business cycle stabi-

lization. We first construct a measure of the tightness gap for the United States (chapter 15) to quantify the scale of the problem. Then, we derive formulas for the optimal conduct of monetary policy (chapter 16) and fiscal policy (chapter 17) based on the tightness gap. We also derive a formula for optimal unemployment insurance based on a generalized tightness gap (chapter 18). The part closes by showing how slack data can be used to detect recessions early (chapter 19).

The book therefore addresses a number of questions that recur during all business cycles and that policymakers systematically face, especially in downturns. It explains for instance how much interest rates should fall when unemployment rises; how large stimulus spending should be in downturns; whether and how the generosity of unemployment insurance should respond to unemployment fluctuations.

Part V concludes the book. We begin by summarizing the main arguments and implications of the slackish framework (chapter 20). We then outline avenues for future research, highlighting how the concept of slack opens up new opportunities for data collection, modeling, and policy design (chapter 21). We also discuss how the models and policy insights developed in the book could be fruitfully applied to countries besides the United States—particularly developing countries, where slack is widespread. We close by situating the slackish business cycle model developed in the book within the macroeconomic landscape—drawing connections to Old Keynesian, Real Business Cycle, and New Keynesian models (chapter 22).

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